Assignment - 2

1)

i \ j				
(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)

यदि i==1 or j==1 or i==n or j==n
तो print(*).

```
int n=5
for (int i=1;i<=n; i++){
        for (int j=1;j<=n;j++){
                if (i==1 || j==1 || i==n || j==n)
                        System.out.print ("* ");
                }else{
                        System.out.print (" ");
                }
        }
        System.out.println ();
}
```

Dryrun : i=1<=5
└ j=1<=5
└ i==1 — true — *
j=2<=5
└ i==1 — ~~false~~ true — *
└ j==1 — false.
j=3<=5
└ i==1 — true *
j=4<=5
└ i==1 — true. *
j=5<=5
└ i==1 — true *

i=2<=5
└ j=1<=5
└ 2==1 — false
└ j==1 — true.
j=2<=5
└ i==1 — false
└ j==1 — false
└ j==n — false
└ i==n — false.

j=3<=5
└ i==1 — false
└ j==1 — f
└ j==n — f
└ i==n — f
j=4<=5
└ i==1 — f
└ j==1 — f

j==n - f
j==n - false.

j=5<=5
- i==1 - f
- j==1 - f
- j==n - true *

i=3<=5
- j=1 — true. *
- j=2 - false.
- j=3 - false
- j=4 false
- j=5 true *

i=4<=5
- j=1 - true *
- j=2 — false
- j=3 - false
- j=4 - false.
- j=5 - true *

i=5<=5
- j=1 - true *
- j=2 - i==n - true *
- j=3 - i==n - true *
- j=4 - i==n - true *
- j=5 - i==n - true. *

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)

n=5.

// if = i==j print(*) or.

// if = i+j = 6 print (*).

```
for (int i=1; i<=n; i++){
    for (int j=1; j<=n; j++){
        if (i==j || i+j==n+1){
            System.out.print("*");
        } else {
            System.out.print(" ");
        }
    }
    System.out.println();
}
```

Dry run :-

$i = 1 <= 5$

- $j = 1 <= 5$
  - $i == j$ $1 == 1$ — true. *
- $j = 2 <= 5$
  - $i == j$ - false
  - $i + j == n + 1$ $(1 + 2) == 6$ - false.
- $j = 3 <= 5$
  - $i == j$ false
  - $1 + 3 = 6$ — false.
- $j = 4 <= 5$
  - $i == j$ - false
  - $1 + 4 = 6$ — false.
- $j = 5 <= 5$
  - $i == j$ - false *
  - $1 + 5 = 6$ - true.

$i = 2 <= 5$

- $j = 1 <= 5$
  - $i == j$ - false
  - $2 + 1 == 6$ - false
- $j = 2 <= 5$
  - $i == j$ - ~~false~~ true. *
  - $2 + 2 == 6$ - false.
- $j = 3 <= 5$
  - $i == j$ - false
  - $3 + 2 == 6$ - false
- $j = 4 <= 5$
  - $i == j$ - false *
  - $4 + 2 == 6$ - true
- $j = 5 <= 5$
  - $i == j$ - false
  - $5 + 2 == 6$ - false

i=3 <=5
└ j=1<=5
└ i==j - false
└ 3+1==6 — false.
j=2<=5
└ i==j - false
└ 3+2==6 — false.
j=3<=5
└ i==j - ~~fa~~ true. *
└ 3+3==6 — true
j=4<=5
└ i==j - false
└ 4+3==6 - false
j=5<=5
└ i==j — false
└ 5+3==6 — false.

i=4<=5
└ j=1<=5
└ i==j — false
└ 1+4==6 - false.
j=2<=5
└ i==j - false *
└ 2+4==6 true
j=3<=5
└ i==j — false
└ 3+4==6 false
j=4<=5
└ i==j — true.
j=5<=5
└ i==j - false
└ 5+4==6 false.

i=5 <=5

└ j=1 <=5

└ i==j = false

i+j ==6 true *

j=2 <=5

└ i==j - false

2+5==6 false

j=3<=5

└ i==j - false

3+5==6 false.

j=4<=5

└ i==j - false

4+5==6 false.

j=5<=5

└ i==j - true. *

Output is :

```
*       *
  *   *
    *
  *   *
*       *
```

	j↓				
i→	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)
	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)
	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)

$j == i$ && $j < i$

Condition is $= j <= i$

```
int n = 5
for ( int i = 1; i<=n ; i++){
        for (int j=1; j<=i ; j++) {
                System.out.print ( "* ");
        }
        System.out.println ();
}
```

Dryrun:

$i = 1$ $1 <= 5$

└ $j = 1 <= 1$ — true *

$j = 2 <= 1$ — false

$j = 3 <= 1$ — false

$j = 4 <= 1$ — false

$j = 5 <= 1$ — false.

$j = 6$ — false. — nextline

$i = 2$ $2 <= 5$ i=

└ $j = 1 <= 2$ — true. *

$j = 2 <= 2$ — true *

j=3<=2 — false.
j=4<=2 — false.
j=5<=2 — false. j=6 — next line

i=3 3<=5
└ j=1<=3 - true *
j=2<=3 — true *
j=3<=3 — true *
j=4<=3 — false.
j=5<=3 - false j=6 next line

i=4 4<=5
└ j=1<=4 - true *
j=2<=4 — true *
j=3 <=4 — true *
j=4 <=4 - true *
j=5<=4 — false
j=6 - false. — next line.

i=5 5<=5
└ j=1<=5 - true *
j=2<=5 — true *
j=3<=5 - true *
j=4<=5 — true *
j=5<=5 - true *
j=6 false i=6 false.

O/P :

```
*
* *
* * *
* * * *
* * * * *
```

To find inverted right triangle we start j from n.

	i j	i j	i j	i
(1,5)	(1,4)	(1,3)	(1,2)	(1,1)
(2,5)	(2,4)	(2,3)	(2,2)	(2,1)
(3,5)	(3,4)	(3,3)	(3,2)	(3,1)
(4,5)	(4,4)	(4,3)	(4,2)	(4,1)
(5,5)	(5,4)	(5,3)	(5,2)	(5,1)

now we see that if i==j && i<j ~~then~~

```
n = 5
for (int i=1; i<=n; i++){
    for (int j=n; j>=1; j--){
        if (i<=j){
            System.out.print("*");
        }else{
            System.out.print(" ");
        }
    }
    System.out.println();
}
```

Dry run :

i=1 1<=5
└ j=5>=1 — *
j=4>=1 — *
j=3>=1 — *
j=2>=1 — *
j=1>=1 — *
j=0 - false. next line

i=2 <=5
└ j=5>=2 — *
j=4>=2 — *
j=3>=2 — *
j=2>=2 — *
j=1>=2 — false. - space print करो.

i=3 <=5
└ j=5>=3 — *
j=4>=3 — *
j=3>=3 — *
j=2>=3 — false — space. else block execute
j=1>=3 - false.

i=4<=5
└ j=5>=4 — *
j=4>=4 — *
j=3>=4 - false
j=2>=4 - false.
j=1>=4 - false

i=5<=5
└ j=5>=5 — *

j = 4 >= 5 - false
j = 3 >= 5 - false
j = 2 >= 5 - false
j = 1 >= 5 - false
j = 0 - false.

i = 6 - false.

and output is :

```
*   *   *   *   *
*   *   *   *
*   **  *
*   *
*
```

*	(1,1)	(1,2) #	(1,3) *	(1,4) #	(1,5)
#	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)
*	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)
#	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)

pattern बघून लक्षात आल जेव्हा i+j हे even number आहेत तेव्हा * print केलय. otherwise print #.

```
int n = 4
for (int i = 1 ; i<=n ; i++) {
    for (int j = 1 ; j<=5 ; j++) {
        int total = i+j ;
        if ( total %2 ==0) {
            System.out.print ("* ");
        }else {
            System.out.print ("# ");
        }
    }
    System.out.println( );
}
```

i+j ची total ला store in variable.
& check if total %2 == 0 means it is even
& print * else print #.

dry run:

i=1 1<=4
└ j=1<=5 — i+j=2 2%2==0
true — *
j=2<=5 — i+j=3 3%2==0 false = #
j=3<=5 — 4 4%2==0 true = *
j=4<=5 — 5 5%2==0 false = #
j=5<=5 — 6 6%2==0 true = *
j=6 = j loop terminate.

i=2 2<=4
└ j=1<=5 — i+j=3 3%2==0 false-#
j=2<=5 — 4 4%2==0 true-*
j=3<=5 — 5 5%2==0 false — #
j=4<=5 — 6 6%2==0 true — *
j=5<=5 — 7 7%2==0 false — #

i=3 3<=4
└ j=1<=5 — i+j=4 4%2==0 true — *
j=2<=5 — 5 5%2==0 false — #
j=3<=5 — 6 6%2==0 true — *
j=4<5 — 7 7%2==0 false — #
j=5<=5 — 8 8%2==0 true — *
j=6 — false.

i=4 4<=4
└ j=1<=5 - i+j=5 5%2==0 false-#
j=2<=5 — 6 6%2==0 true — *
j=3<=5 — 7 7%2==0 false — #
j=4<=5 — 8 8%2==0 true — *
j=5<=5 — 9 9%2==0 false — #

o/p :

```
*  #  *  #  *
#  *  #  *  #
*  #  *  #  *
#  *  #  *  #
```